

Worksheet 5 *Power functions*

Prepares you for lab manual pages **Exp5, Exp6, Exp7, Exp8** (section L2 S4)

Name:

Name of TA:

1. Suppose that $F(t) = 9t^{2/3}$.

(a) Draw a time series for the quantity F . (Use Desmos to help if you wish.)



(b) What is $F(8)$? (Give the exact value; it is a whole number.)

(c) What is $F(8.01)$? (Use a calculator.)

(d) What is ΔF on the time interval from $t = 8$ to $t = 8.01$?

(e) Estimate dF/dt when $t = 8$.

2. A circular spot of mold is growing on the ceiling. Its radius is $R(t) = 2t$, where t is time in days and R is in millimeters.

(a) Describe the area of the mold spot as a function of time.

$A(t) =$

(b) Estimate dA/dt when $t = 3$.

$dA/dt \approx$

3. Carefully draw a tangent line with a ruler, and use it to estimate dY/dt when $t = 3$.



$dY/dt =$

when $t = 3$.